

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Computer Science and Engineering

VolcanoX - A Real-Time Volcanic Eruption Simulation in OpenGL

Lab Project Proposal

Course Title: Computer Graphics & Image Processing Laboratory

Course No: CSE 4102

Submitted By
Shormi Ghosh
Roll: 2107109

Submitted To
Md Tajmilur Rahman
Lecturer
Md. Mubtashim Abrar Nihal
Lecturer
Dept. of CSE, KUET

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Project Description:

Overview - The scene will be built using only simple shapes such as cones, spheres, discs, and small blocks. No textures or pre-made models will be used. All motion will be calculated every frame using elapsed program time. Nothing will be pre-recorded, so every object's position and size will be explainable by a formula at any moment.

Scene Construction - The volcano will be a cone made from a ring of points around a peak. Small rocks around the crater will be drawn in one efficient batch. Lava droplets, rock chunks, and ash puffs will use fixed reusable pools. When an object finishes its motion, it will be reused instead of being created or destroyed during runtime.

Movement - Lava droplets will launch from the crater at regular intervals and will follow projectile motion using initial velocity and gravity, while slightly cooling and shrinking during flight. Rock chunks will follow the same arc while continuously spinning. The lava flow will use a moving brightness pulse along a fixed strip instead of moving the geometry. The lava pool will slowly grow and level off. The crater glow will pulse in size and brightness. Ash puffs will move upward, expand, fade, and then be reused.

Transformations - All three basic transformations will be used:

- **Translation:** will move lava, rocks, and ash.
- **Rotation:** will spin rock chunks and position rim rocks around the crater.
- **Scaling:** will change ash size, shrink cooling lava, grow the lava pool, and set the size of basic shapes.

Multiple transformations will be combined in many objects. For example, a flying rock will be translated along its path while rotating around its own axis. The rim rocks will also demonstrate hierarchical transformation, as their positions will depend on the crater. Therefore, moving the crater will move all the rocks with it.

Lighting - Two lights will be used. The crater will act as the main point light, becoming weaker with distance. A dim directional moonlight will provide background lighting. Surface brightness will use three components: ambient, diffuse and specular lighting.

Material Contrast - Rock and lava will use the same lights but different material settings. Rock will have low shininess and a weak highlight, giving it a matte appearance. Lava will have high shininess and a strong specular highlight, making it look wet and glossy. Lava will also have a small glow to represent its heat.

Shading Comparison - The project will support switching between Flat, Gouraud, and Phong shading. Flat shading will light each triangle once. Gouraud shading will calculate lighting at the triangle's corners and blend it, which may miss highlights in the middle. Phong shading will interpolate the surface direction and calculate lighting at each point, producing more accurate highlights. This will be demonstrated on the lava pool, where Phong may preserve a highlight that Gouraud can lose at low polygon counts.

Efficiency - The project will reuse a small set of shapes, use fixed pools for droplets, rocks, and ash, and skip inactive objects. Repetitive calculations such as the rim-rock ring and glow will be handled efficiently by the graphics card. No textures, shadows, or reflections will be used, keeping the focus on transformations, animation, lighting, and shading.